

Assignment 1 of MATP4820

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Problem 1

Let the matrix $A = \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 3 \end{bmatrix}$. Compute $\|A\|_1$, $\|A\|_2$, and $\|A\|_\infty$.

matrix 1-norm: $\|A\|_1 = \max_j \sum_i |x_{ij}| = \max\{(|2| + |0|), (|-1| + |-3|), (|2| + |3|)\} = 5$

matrix 2-norm:

First computing AA^T :

$$AA^T = \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & -3 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 2(2) - 1(-1) + 2(2) & 2(0) - 1(-3) + 2(3) \\ 0(2) - 3(-1) + 3(2) & 0(0) - 3(-3) + 3(3) \end{bmatrix} = \begin{bmatrix} 9 & 9 \\ 9 & 18 \end{bmatrix}$$

Now computing eigenvalues of AA^T :

$$\det(\lambda I - AA^T) = \det \begin{bmatrix} \lambda - 9 & 9 \\ 9 & \lambda - 18 \end{bmatrix} = (\lambda - 9)(\lambda - 18) - 81 = 0$$

Solving for λ :

$$\lambda^2 - 27\lambda + 81 = 0 \implies \lambda = \frac{27 \pm \sqrt{27^2 - 4(81)}}{2}$$

Therefore, taking largest eigenvalue, $\|A\|_2 = \sqrt{\frac{27 + \sqrt{405}}{2}} = \sqrt{\frac{27 + 9\sqrt{5}}{2}}$

matrix ∞ -norm: $\|A\|_{\infty} = \max_i \sum_j |x_{ij}| = \max\{(|2| + |-1| + |2|), (|0| + |-3| + |3|)\} = \mathbf{6}$

Problem 2

Compute the gradient and Hessian matrix of the function $f(x_1, x_2) = 25(x_2 - x_1^2)^2 + (1 - x_1)^2$. Find all local minimizers of this function. Justify your answer.

First expanding $f(x_1, x_2)$:

$$f(x_1, x_2) = 25(x_2^2 - 2x_1^2x_2 + x_1^4) + (1 - 2x_1 + x_1^2) = 25x_1^4 + 25x_2^2 - 50x_1^2x_2 + x_1^2 - 2x_1 + 1$$

Compute the gradient:

$$\nabla f(x_1, x_2) = \begin{bmatrix} 100x_1^3 - 100x_1x_2 + 2x_1 - 2 \\ 50x_2 - 50x_1^2 \end{bmatrix}$$

Compute the Hessian Matrix:

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 300x_1^2 - 100x_2 + 2 & -100x_1 \\ -100x_1 & 50 \end{bmatrix}$$

Now solving $(x_1, x_2) = 0$:

$$\begin{cases} 100x_1^3 - 100x_1x_2 + 2x_1 - 2 = 0 \\ 50x_2 - 50x_1^2 = 0 \end{cases}$$

Solving for x_1, x_2

$$\implies x_1 = 1 \implies x_2 = 1$$

So the solution is $x_1 = 1, x_2 = 1$, and this point is a stationary point.

Now checking the Hessian Matrix at stationary point (1,1):

$$\nabla^2 f(1,1) = \begin{bmatrix} 202 & -100 \\ -100 & 50 \end{bmatrix}$$

$$\lambda_1 + \lambda_2 = 202 + 50 = 252$$

$$\lambda_1 \lambda_2 = 202(50) - (-100(-100)) = 100$$

$$\implies \lambda_1 > 0, \lambda_2 > 0$$

Thus the Hessian Matrix is positive definite, so $x_1 = 1, x_2 = 1$, or (1,1) is the a local minimizer, and the only local minimizer of this function.

Problem 3

Let $f(x_1, x_2) = \frac{1}{2}x_1^2 + cx_1x_2 + x_2^2 - x_1 + 2x_2 - 4$, where c is a real number. Give all values of c such that f is convex.

We will use the known fact that $f(x_1, x_2)$ is a twice differentiable function, and that if $\nabla^2 f(x)$ is Positive Semi-definite at any x , then f is convex.

First, computing the gradient:

$$\nabla f(x_1, x_2) = \begin{bmatrix} x_1 + cx_2 - 1 \\ cx_1 + 2x_2 + 2 \end{bmatrix}$$

Now Hessian:

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 1 & c \\ c & 2 \end{bmatrix}$$

Since $1 \geq 0$ and $2 \geq 0$, we just need $1(2) - c^2 \geq 0$:

$$2 - c^2 \geq 0 \implies 2 \geq c^2 \implies -\sqrt{2} \leq c \leq \sqrt{2}$$

So, with $-\sqrt{2} \leq c \leq \sqrt{2}$ our Hessian matrix will be Positive Semi-definite. With the Hessian Positive Semidefinite, f will be a convex function. Thus, f is a convex function for $-\sqrt{2} \leq c \leq \sqrt{2}$.

Problem 4

Is the sequence $x^{(k)} = 1 + (-\frac{3}{5})^{2^k}$ for $k = 1, 2, \dots$, Q-quadratically convergent to a finite real number? If yes, give the limit and prove it; if not, explain why.

Yes, the sequence is Q-quadratically convergent to the finite limit $x^* = 1$

Proof. Suppose $x^{(k)} \rightarrow x^*$ as $k \rightarrow \infty$. We say the convergence is Q-quadratic if there is $M > 0$ such that

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} \leq M, \text{ where } k \text{ is large enough}$$

First, we can find x^* :

$$\left(-\frac{3}{5}\right)^{2^k} \rightarrow 0 \text{ as } k \rightarrow \infty$$

So,

$$x^{(k)} \rightarrow 1 \text{ as } k \rightarrow \infty. \text{ Take } x^* = 1.$$

Now,

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} = \frac{\left|1 + \left(-\frac{3}{5}\right)^{2^{k+1}} - 1\right|}{\left|1 + \left(-\frac{3}{5}\right)^{2^k} - 1\right|^2} = \frac{\left(\frac{3}{5}\right)^{2^{k+1}}}{\left(\frac{3}{5}\right)^{2 \cdot 2^k}} = \frac{\left(\frac{3}{5}\right)^{2 \cdot 2^k}}{\left(\frac{3}{5}\right)^{2 \cdot 2^k}} = 1.$$

Thus,

$$\frac{\|x^{(k+1)} - x^*\|_2}{\|x^{(k)} - x^*\|_2^2} \leq M, \text{ with } M = 1 > 0.$$

Therefore, we have shown that the sequence is Q-quadratically convergent to $x^* = 1$

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